

2024

nimbus

THERAPEUTICS

Accelerating the Design of Candidate Drugs Using Amazon SageMaker with Nimbus Therapeutics

Learn how computational chemists at Nimbus Therapeutics built an agile and effective ML pipeline using Amazon SageMaker.

[Overview](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

Saved significant time

with automated ML model deployment

Less than 1 second

to view predictions of a molecule's viability

Invested drug discovery resources

in most promising ideas



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Overview

With established expertise in computational chemistry and machine learning (ML), [Nimbus Therapeutics](#) (Nimbus) sought to further accelerate its drug discovery engine by automating the operational aspects of ML (MLOps). Minimizing manual intervention in model training and deployment would help its scientists to focus on Nimbus's core mission: the design of breakthrough medicines.

Nimbus turned to Amazon Web Services (AWS) to use [Amazon SageMaker](#), which lets organizations build, train, and deploy ML models for any use case with fully managed infrastructure, tools, and workflows. Using Amazon SageMaker as part of its technology stack, Nimbus built an MLOps pipeline that automates the deployment of predictive models, which scientists use to design molecules with improved drug-like properties before synthesis and testing. As a result, scientists can allocate precious resources to ideas that are most likely to succeed.

Furthermore, the automation frees the computational chemists to do what they do best. "If we are constantly training models step by step, it takes us away from designing drugs," says Leela Dodda, director of Computational Chemistry at Nimbus.



Opportunity | Gaining Insight into a Molecule's Viability in Milliseconds

Founded in 2009, Nimbus designs breakthrough medicines by creating potent, selective small molecules that target proteins that are known to be fundamental drivers of cancer, immune conditions, and metabolic diseases. One of Nimbus's core principles is the use of protein structure and molecular simulation to guide the design of new



molecules toward greater efficacy and selectivity. And while physics-based approaches are foundational, many biological processes that determine a drug's safety and effectiveness, such as drug absorption or drug metabolism, are not easily modeled with physics. Modern ML offers a complementary approach to predicting costly experimental outcomes and biasing molecular design toward more drug-like properties by learning historic experimental trends, which are published in scientific literature, patent applications, and proprietary datasets.

"It's expensive and time-consuming to synthesize and test compounds," says Dan Price, vice president of Computational Chemistry and Structural Biology at Nimbus. "The intent is to predict how molecules could behave in the body at the time of design and use that near real-time feedback to redesign toward better properties before synthesizing each molecule."

Solution | Using Amazon SageMaker to Save Time and Resources in Drug Design

The team at Nimbus imagined an automated pipeline that would regularly curate data from a variety of sources and retrain and redeploy ML models. The pipeline would provide near real-time predictions through Schrödinger's LiveDesign, an application where new molecular designs are captured and evaluated. Many data sources in the pharmaceutical domain are small, inherently heterogenous, and missing key structured metadata. Nimbus considered it critical to codify domain-expert rules for metadata extraction and criteria for inclusion and exclusion to sustainably deliver performant models.

The team needed to quickly demonstrate whether the available data, combined with its curation and modeling strategies, could meaningfully impact programs and warrant a larger investment in time and resources. For prototyping, the team employed [Amazon SageMaker notebooks](#), which are fully managed notebooks in JupyterLab for exploring data and building ML models with automatic scaling of compute resources. This generative artificial intelligence coding companion aided in authoring Amazon SageMaker-specific code. To orchestrate communication with LiveDesign, the team used [Amazon API Gateway](#), a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale. Amazon API Gateway used the serverless functionality of [AWS Lambda](#), which lets organizations run code without thinking about servers or clusters, to provide near real-time inference from Amazon SageMaker endpoints. This integrated solution, although not fully automated, validated the team's approach.

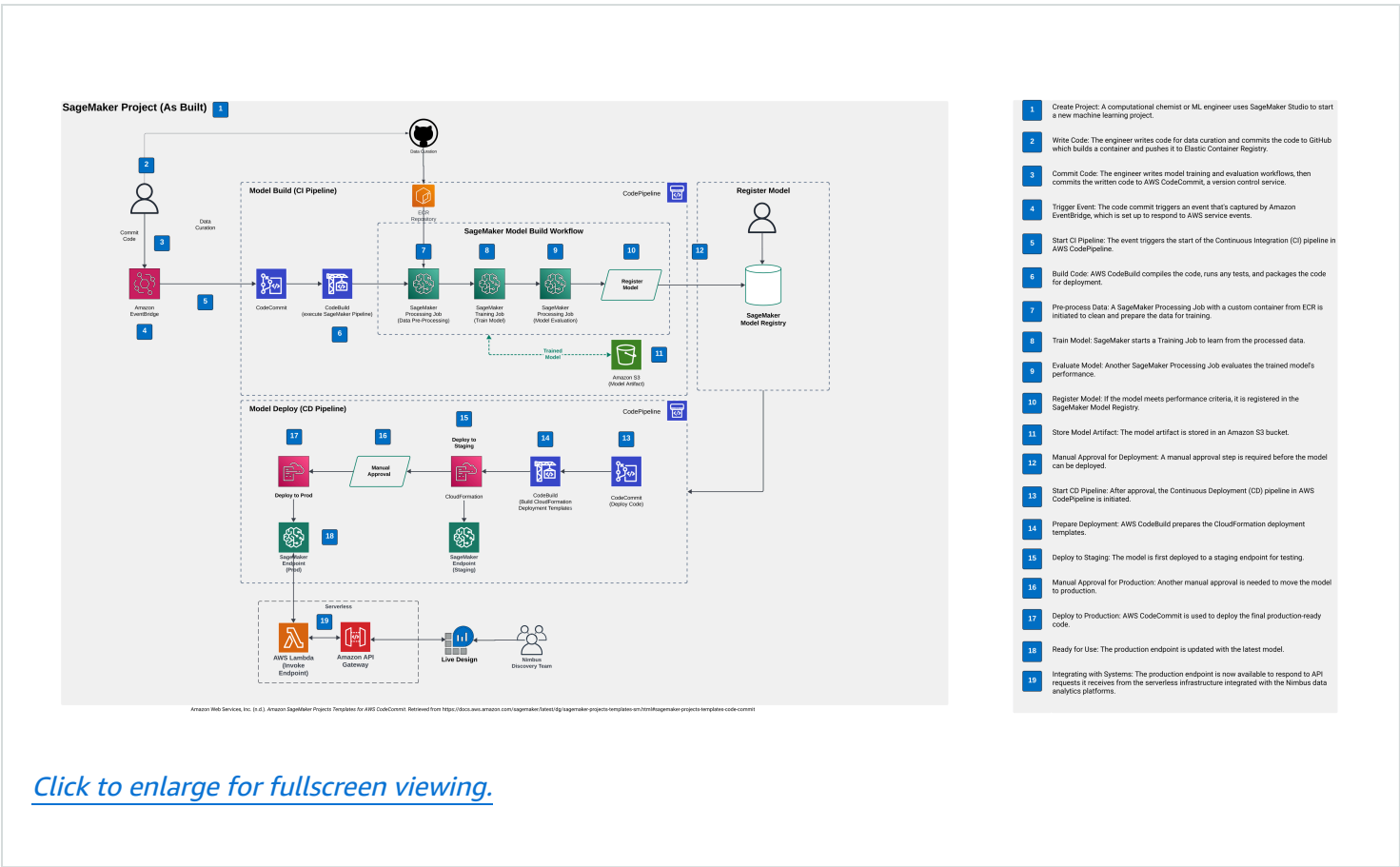
Working with their own IT team and with AWS specialists, Nimbus scientists built on this foundation by automating the pilot with [Amazon SageMaker Pipelines](#), a serverless workflow orchestration service purpose-built for ML automation. Subsequent migration to [Amazon SageMaker Projects](#), which offers continuous in-



delivery systems for MLOps engineers, provided a sustainable solution for continuous implementation of improvements and scheduled retraining.

“Our job is to deliver drug candidates,” says Price. “Any technology needs to be fully aligned with that mission and not take a life of its own or become a distraction. AWS addresses this exact challenge: giving us the tools to develop a custom, focused solution without having to develop an entire DevOps or MLOps service ourselves.”

Architecture Diagram

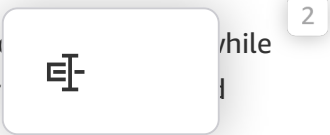


Outcome | Expanding the Automated ML Pipeline to Bring Therapies to Market Faster

Nimbus has successfully automated a suite of ML models to predict different aspects of drug exposure and safety, which are actively influencing design across all discovery projects. The predictions provided through LiveDesign democratize the technology across all Nimbus scientists, regardless of discipline or technical background. However, the infrastructure isn't specific to LiveDesign. In fact, Nimbus is now using iterative rounds of ML predictions for reinforcement learning, guiding molecular generative artificial intelligence to explore more chemical space and inject novel design concepts into the team's thinking.

The engineering framework developed for MLOps has proven valuable beyond its initial scope. It can be adapted to scale various types of scientific computing, including tasks in chemoinformatics and physics. Nimbus continues to identify opportunities to use this solution, aiming to streamline computational sciences and enhance process efficiency. “One advantage of investing in this solution is that it has spawned many different ideas,” says Dodda. “We can use the same technology to tackle other problems.”

“Using Amazon SageMaker helps us bring more science to identifying the most efficacious and while trimming the time from program inception to the clinic,” says Price. “That’s important to ever speed matter.”



About Nimbus Therapeutics

Founded in 2009, Nimbus Therapeutics uses computational technology to drive drug discovery in cancer, autoimmune conditions, and metabolic diseases.

AWS Services Used

Amazon SageMaker

The next generation of Amazon SageMaker is the center for all your data, analytics, and AI.

[Learn more »](#)

Amazon SageMaker notebooks

You can use SageMaker notebook jobs to create a non-interactive job to either run on demand or on a schedule. Use an intuitive user interface or SageMaker Python SDK to schedule your jobs right from JupyterLab.

[Learn more »](#)

Amazon SageMaker Pipelines

Amazon SageMaker Pipelines is a serverless workflow orchestration service purpose-built for MLOps and LLMOps automation. You can easily build, execute, and monitor repeatable end-to-end ML workflows with an intuitive drag-and-drop UI or the Python SDK.

[Learn more »](#)

Amazon SageMaker Projects

SageMaker Projects help organizations set up and standardize developer environments for ML and CI/CD systems for MLOps engineers.



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